Which Value Is Still in Force? Supersession-Aware Retrieval over Corrected Clinical Records

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***Abstract*—Clinical charts only ever grow. When a lab value was wrong, a corrected result gets filed beside it and the old entry stays, since a doctor may already have acted on it. Ask a similarity-based retrieval assistant for that value and it has no idea which entry still counts. The two look almost identical to an encoder, and neither says it was withdrawn. We call this the supersession problem. In SoloPrac, a record and question-answering system we built for single-doctor clinics in India, a correction is stored as a new event that points at the event it replaces, and the state of every event is worked out on read for a given query instant. A value corrected in March is therefore current for a question about February and superseded for a question about April. To test the idea we wrote TAP-QA, a synthetic benchmark with about 275 patients, 4,500 events and 850 queries per seed, in which the right answer to every corrected and as-of question is known by construction. On two seeds our ranker never returns a withdrawn value first (0.0% stale) and gets 97.8% of historical questions right. A ranker that sees valid time and transaction time, but not the pointer, gives 7.0% and 11.5% stale answers and stays under 8% on historical questions. None of 35 recency-decay settings answered a single historical question correctly, and a plain as-of filter on the same pointers did as well as our weighted rule, so the gain comes from keeping the relation at all.**

***Index Terms*—retrieval-augmented generation, electronic health records, temporal databases, data provenance, benchmark, de-identification**

I. Introduction

Suppose a patient's serum creatinine comes back at 1.1 mg/dL on the 3rd of the month. On the 6th the laboratory re-runs the specimen and reports 1.4 mg/dL. Both numbers now sit in the chart. Deleting the first is not allowed (a prescriber may have dosed a drug on it), and quietly overwriting it would hide the fact that the advice changed. A clinician reading the chart has no trouble with this, because the second entry says "corrected result" and refers back to the first.

A retrieval pipeline has a lot of trouble with it. The usual design embeds every entry, compares it with the question and passes the top few entries to a language model. For the question "what is the serum creatinine?" the two entries get almost the same score. In our benchmark they very often get exactly the same score, since they share the test name and differ only in a number and a status word. Whichever one wins comes back with a valid citation, and when the old one wins the doctor reads a withdrawn value presented as fact.

We ran into this while building SoloPrac, a record system for Indian doctors who run a clinic alone, with no IT staff. That gave us hard limits: one laptop, zero monthly cost, and no identifiable data on anybody else's server (the Digital Personal Data Protection Act, 2023, is strict about that). The failure is easy to show on our own data: with plain dense retrieval, about half the questions about a corrected test put the withdrawn value first (49.5% on TAP-QA).

A result typed in late for an earlier date only needs a second clock. A retraction is harder: both values were believed when written, and only the reference inside the correction shows that the first stopped holding. Ranking by recency gets the present-tense question right by accident and every question about the past wrong. Clinical standards have stored the reference for years through HL7 result status codes, FHIR Observation status and Provenance, and openEHR versioning [4], [5]. Recent memory stores for LLM agents keep something very similar [6]–[9]. Nobody, as far as we could find, had measured how much a clinical retriever gives up by discarding that reference when it must also answer questions about the past. Our contributions are:

- a formulation in which the state of a record depends on the record and on the query instant, computed over the supersession graph, with an as-of rule that keeps the past answerable (Sections III and IV);
- TAP-QA, a generator and harness with four query classes, distractor corrections and ground truth by construction, which calls the retriever that ships in SoloPrac (Section V);
- results on two seeds with patient-clustered intervals and paired tests, plus two negative findings: no recency prior

reaches the useful region, and a hard as-of filter on the same edges matches our weighted rule (Section VI);
- a working clinic system built around the ranker (Section IV).

II. Related Work

Time in databases and clinical standards. Database people have had two clocks for decades. Snodgrass and Ahn [1] called them valid time and transaction time (roughly: when it was true, and when we wrote it down), and the glossary of Jensen et al. [2] pinned the words down. A bitemporal row carries both times but still cannot say that a later row cancelled it. That kind of link does have a name in W3C PROV [3], where one entity can be a revision of another. Clinical formats store it too. In FHIR you can set an Observation's status to `corrected` or `entered-in-error` and point a Provenance resource at the old one [4]; openEHR goes further and versions everything, each version knowing its parent [5]. So hospitals often do have the link. It gets lost later, somewhere between chunking and the vector index.

Agent memory and versioned documents. People building memory for LLM agents arrived at much the same idea we did. Zep [6] keeps bitemporal edges in a knowledge graph and marks old edges invalid instead of deleting them. MemStrata [7] and Engram [8] both keep a pointer from new fact to old and filter on it before dense ranking, and MemStrata gets stale-fact rates near zero on software configuration data. SodaMem [9] types its edges, VersionRAG [10] builds version graphs over documentation, and TempRALM [11] mixes a closeness-in-time term into its ranking. None of them, as far as we can tell, was tested on clinical records or reports historical accuracy next to staleness.

Time-aware ranking and outdated evidence. Campos et al. [12] note that a query's time is hard to detect; we sidestep that by supplying it. For the baselines without any notion of time we used dense retrieval [13] with MedCPT [14], BM25-style token overlap [15] and reciprocal rank fusion [16]. HoH [17] reports that an outdated passage does more damage to a RAG answer than an irrelevant one. Ahmed et al. [18] built a set of 317 knowledge reversals (some medical) and found outdated retrieval flipping roughly one answer in three, with a recency re-ranker recovering only a few points of that. Chronofy [19] instead learns decay coefficients from evidence age, which is the family we put to the test in Section VI-D.

Medical RAG benchmarks. MIRAGE [20] and RGB [21] test whether the system can find a passage; nothing in their corpora is out of date. ChronoMedKG [22] asks about disease onset. MEDEC [23] hands a model a note with a planted error. None of them contains a value that was correct when entered and withdrawn later, which is the case we care about.

III. Problem Formulation

Take one patient and call the log E. Every event e in it has a payload and the two clocks from above, $v(e)$ and $\tau(e)$. Corrections have one more thing: a reference, never edited afterwards, to the event they replace. We write $c \rightsquigarrow e$ when c corrects e.

Timestamps cannot recover $\rightsquigarrow$. Take the creatinine pair again. By the 7th, say, both creatinine entries lie in the past on both clocks. Filter on either clock and you still have both. Only the pointer in c marks e as void.

Dating a correction. We set $v(c)$ to the moment the corrected value came into force (the re-run on the 6th) and keep the specimen date on the original. If both versions carried the specimen date, "what was the creatinine on the 4th?" would have two answers. Our rule answers "what did the chart say at t?", the auditor's question. The re-analyst's question, "what do we now believe was true at t?", needs back-dated corrections, which we did not evaluate.

State at an instant. For a query instant t,

$$\mathrm{state}(e,t) = \begin{cases} \text{superseded} & \text{if } \exists\, c:\ c \rightsquigarrow e \ \wedge\ v(c) \le t, \\ \text{current} & \text{otherwise.} \end{cases} \tag{1}$$

Most of the clinical sense sits in the test $v(c) \le t$. Without it a record corrected in March looks superseded for a question about February, and the ranker answers today's questions well and every older one badly. Our own first harness had exactly this defect (Section V). TAP-QA hands t_q and the intent to every method as given facts, which keeps date parsing out of the measurement.

Metrics. The stale answer rate (SAR) is the share of corrected-class queries whose top-ranked record is superseded at t_q. For historical accuracy we look at as-of queries only, and check whether the first record back was the one on file on the date in question. Watch out for a cheap trick here. Always returning the newest entry gives a perfect SAR of zero, and also zero historical accuracy. That is why we add a safe-answer rate, which needs the top record to be both the gold one and not superseded.

Statistics. Queries about the same patient are correlated, so all bootstrap intervals resample patients with replacement (10,000 resamples), and McNemar's exact test [24] runs on paired per-query outcomes. Against the bitemporal baseline the gain in historical accuracy is $+0.9125$, with a patient-clustered 95% interval of $[+0.875, +0.948]$.

IV. System and Method

A. SoloPrac in brief. The doctor uses SoloPrac in a browser or as an installed phone app (we wrote it in React). The server side is FastAPI talking to PostgreSQL, which holds the event log, plus Qdrant for vectors, Redis for queues and SeaweedFS for files. We run all of it, seven containers in total, on one student laptop with an i5-12450H, 16 GB of RAM and a 4 GB RTX 3050. Doctors add and correct records, ask questions, and issue prescriptions and certificates as PDFs to the patient's inbox. There is no status column on an event. A correction is an `INSERT` whose `supersedes_id` names its predecessor, and a database check constraint rejects a pointer to another patient's event, a correction dated before its predecessor, and cycles.

B. Supersession graph. When a patient's log is loaded, the pointers are collected into a per-patient DAG. Chains are allowed and state passes along them transitively. After the build, a state lookup takes constant time. Construction raises an error if a correction points at a missing event or at another patient's event. It can always be rebuilt from the log.

C. Using the edge. With $\mathrm{state}(e, t_q)$ from (1), the simplest option is a hard filter: drop every record that is not admissible at t_q and rank what is left by similarity. SoloPrac ships a soft version. For a query q with intent i,

$$\mathrm{score}(q, e) = \alpha\, \mathrm{sem}(q, e) + (1 - \alpha)\, w(e, i), \qquad (2)$$

with $\alpha = 0.70$. Here sem is the cosine similarity from MedCPT [14], with token overlap as a fallback when the encoder or the index is down. For w we only needed three numbers. A record in force at the moment the question asks about gets 1.00. For an undated question, current records get 0.85 and superseded ones 0.15. One consequence is worth spelling out: the corrected HbA1c of last spring scores the full 1.00 if you ask what it was last spring, because back then it was the value on file. Superseded records are pushed down the list and kept in it, so a reader asking what the chart said at the time sees the withdrawn value beside its correction. We tried seven values of α between 0.40 and 0.95 on seed 20260601, and SAR, corrected accuracy and historical accuracy did not move at all. The weight term spans 0.15 to 1.00 while one patient's similarity scores sit in a narrow band, so the order inside a record-correction pair never flips.

D. Tiebreak and veto. For an as-of question every admissible record gets the same weight, and a second key has to settle the order. Sorting by recency would return the newest admissible record, which is often the wrong one, so we sort by the distance between $v(e)$ and t_q. On seed 20260601 the scores tie on 161 of the 850 queries and this key decides them. When the intent carries an instant, anything dated after t_q is thrown out before scoring. A summary of the past must not quote the future.

E. Encoders and chunking. Each event is one retrieval unit. We flatten it into a short text: a type label ("Laboratory result", "Medication" and so on) followed by its payload fields as `key: value` pairs, cut at 1,800 characters so it fits inside MedCPT's 512-token window. An event usually holds one fact; splitting it would cut a value off from its unit, and merging events would mix validity states in one vector. The MedCPT article encoder embeds the records and the separate query encoder embeds the question; each tower is a BERT-base model (12 layers, 768 dimensions, about 110M parameters) trained contrastively on 255M PubMed query-article pairs [14]. Token vectors are mean-pooled over the attention mask and L2-normalised. Token overlap scores "what is her blood sugar" at 0.000 against a fasting glucose record, while MedCPT gives 0.825 and ranks that record first. Images go through BiomedCLIP [25], a ViT-B/16 image tower with a PubMedBERT text tower, 512 dimensions, trained on 15M figure-caption pairs from PubMed Central. On four labelled test images each one was closest to its own caption (median margin 0.255). The patient filter sits inside the Qdrant query itself.

F. De-identified generation and provenance. Before any text leaves the machine, names, phone numbers, emails, Aadhaar-shaped numbers, record numbers and long digit runs are swapped for typed placeholders, and calendar dates become offsets from today (`[T_MINUS_14D]`), in line with the surrogate approach common in clinical de-identification [26]. Record identifiers become R1, R2 and so on, and are mapped back locally. A last regex gate runs on the exact outbound string and blocks the request if anything identifiable is left. The model's answer is split into claims. Each claim has to cite a retrieved record, the numbers it states have to match that record, and a present-tense claim may not cite a superseded record. Claims that fail are dropped, and if all of them fail the doctor gets a cited extract of the records with no prose at all. Photographed lab reports go through RapidOCR, and a read with confidence below 0.75 is held for the doctor to confirm; it never enters the record on its own.

V. The TAP-QA Benchmark

Seed 20260601 produces 277 patients and 4,500 events: 3,251 laboratory results, 489 diagnoses, 446 medications and 314 notes; 972 of these events are corrections. There are 850 queries. 200 are *unchanged* (no correction touches the test), 200 are *corrected* (the gold record is the correction), 400 are *historical* (as-of questions, answered by whichever version held on that date) and 50 are *contradiction* probes that should return two clashing records together. Seed 20260714 is similar (274 patients, 1,001 corrections, 852 queries). A correction moves the original value by 12 to 55% of the test's reference range and is filed 1, 3 or 14 days later, and the lag is skewed on purpose, so neither the size of the change nor its age gives a correction away. Every patient also gets distractor corrections on tests no query asks about, and a method that simply demotes every correction pays for it.

The data is synthetic on purpose. In real charts the correction is nearly always the newest entry for its test, which is exactly what makes recency look like a fix; generating data breaks that link (see Section VII and [27]).

The main table scores rankings. Generated text varies from run to run, and when an end-to-end answer is wrong you cannot tell whether retrieval was to blame. To make sure we tested the real thing, the harness builds an actual supersession graph and calls SoloPrac's own `ValidityAwareRetriever` through a thin adapter. We learned this the hard way. Our first harness had a hand-written copy of the method that asked "is this record superseded *now*?" instead of "was it current at the asked instant?", and it put our method at 28% historical accuracy while the app itself got those same questions right.

Ties needed care as well. Suppose a method gives a record and its correction the very same score. Then it has picked neither, and we compute rank from each method's own scores with ties averaged; two records tied at the top each get rank 1.5. Our first harness broke ties by record ID, which follows insertion order and hence time, and token overlap appeared

TABLE I
RETRIEVAL ON TAP-QA, TWO SEEDS OF ABOUT 850 QUERIES EACH (%). SAR OVER 200 CORRECTED QUERIES, HIST. OVER 400 AS-OF QUERIES, SAFE = GOLD RECORD FIRST AND NOT SUPERSEDED.

	SAR ↓		Hist. ↑		Safe ↑	
Method	s1	s2	s1	s2	s1	s2
Random floor	17.0	22.5	9.2	7.2	4.5	5.5
Token overlap	100.0	100.0	0.0	0.0	0.0	0.0
Dense only (MedCPT)	49.5	53.5	45.8	53.0	50.5	46.0
RRF, no temporal term	69.5	73.0	55.0	60.5	1.5	1.5
Valid-time only	0.0	0.0	0.0	0.0	100.0	99.5
Transaction-time only	21.5	24.0	21.2	23.2	78.5	75.5
Latest-N	0.0	0.0	0.0	0.0	10.5	11.5
Bitemporal (both axes)	7.0	11.5	6.5	7.5	87.5	83.0
Graph filter (as-of)	0.0	0.0	98.2	98.2	100.0	99.5
SoloPrac (ours)	**0.0**	**0.0**	**97.8**	**97.8**	**98.5**	**96.0**

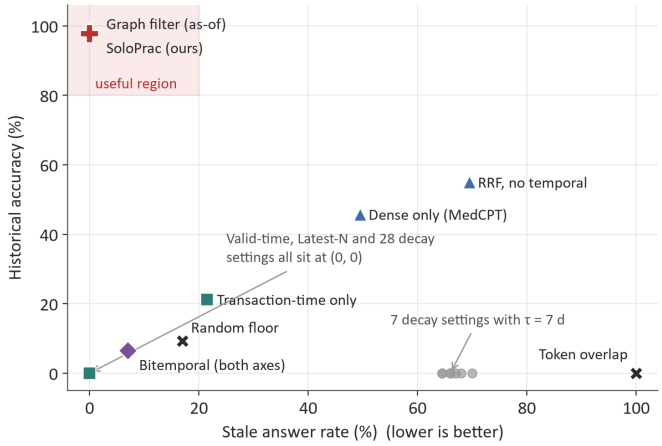


Fig. 1. Stale answer rate against historical accuracy for every configuration we ran on seed 20260601. The graph filter and our ranker overlap near the top left. The decay settings all end up on the bottom axis.

to reach 99.8% historical accuracy. Read correctly, it scores 0.0%.

One command, `python -m tapqa.run --seed S`, regenerates everything below: the ten methods of Table I, a 35-point decay grid, and 10,000 bootstrap resamples. If you run it twice you get the same file back, except for timestamps and timing fields.

VI. RESULTS

A. Main comparison

Table I lists the ten methods on both seeds. Fig. 1 plots all 45 configurations of seed 20260601 by SAR and historical accuracy.

Without the graph, no method does well on both axes. The timestamp rankers fail one or the other, and Latest-N fails both. Valid-time only adds $0.30\exp(-\text{age}_v/90)$ to similarity, and its 0.0% SAR is a side effect of our dating: the correction is always dated later, so this ranker picks it by luck, and it has no way to answer as-of questions. When we re-dated every correction to its specimen, valid-time only jumped to 100% SAR (it had been 0%) and bitemporal to 70% (it had been 7%). Our SAR stayed at 0%, because the graph never compares timestamps. Our historical accuracy did drop to zero under that convention, as the dating rule of Section III implies.

B. Graph filter against the weighted rule

The graph-filter row uses the same edges, keeps only records admissible at t_q and orders the survivors by MedCPT similarity alone. It did at least as well as our rule: 98.2% against 97.8% historical accuracy on both seeds, and 100.0/99.5% against 98.5/96.0% on the safe-answer rate (McNemar $p = 0.25$ on seed 1, and $p = 0.016$ in the filter's favour on seed 2). Where nothing was corrected, the weighting costs two or three points (94.5/96.5% against 97.5/98.0% on unchanged queries). (Every method gets 100% on the contradiction probes.) We still ship the soft rule because it keeps the correction visible to the model, a design reason we have not measured.

C. Stability and significance

Our ranker returns 0 stale answers out of 200 on each seed, and also under the re-dating counterfactual. With 0/200 the 95% upper bound on SAR is about 1.9%. On historical questions it got 391 of 400 right, on each seed. Baselines move between seeds (bitemporal SAR by 4.5 points); our headline numbers do not. For the safe-answer outcome on seed 1, McNemar's test gives $p = 2.7 \times 10^{-5}$ against bitemporal and 8.0×10^{-27} against dense-only.

D. Recency decay

The obvious objection to all of this is that a recency term, tuned well, might do the same job. So we tried it: $f(q, e) = \text{sem}(q, e) + \lambda \exp(-\text{age}(e)/\tau)$ with seven values of λ from 0.01 to 0.5 and τ of 7, 30, 90, 180 and 365 days. That is 35 settings per seed, and none of the 70 got a single historical question right. Once we printed the scores the reason was obvious. A test and its correction have the same name and almost the same payload, and on every corrected and historical query of both seeds the similarity term scored the pair exactly equal (we checked all 600 per seed). So the decay term alone picks the winner. It favours the newer record, which is right for corrected queries and, by construction, wrong for every historical one. For $\tau \geq 30$ days the outcome is 0% SAR and 0% historical accuracy everywhere on the grid. The $\tau = 7$ row is odd for a different reason. Record ages go from 30 up to 435 days, and $\exp(-\text{age}/7)$ over that range shrinks by around 25 orders of magnitude. The two versions then differ by less than a double can represent, they tie, our tie rule credits neither, and SAR comes out at 64.5 to 70.0% on seed 1. This is one family of priors, and we make no claim about other families.

E. Effect on generated answers

Table I tells us whether the right record comes first. To see whether that changes the answer, we took 40 corrected queries and 40 unchanged controls and had the same hosted model answer each one twice: once with the context our ranker picks and once with the context dense-only picks. Neither arm marked validity. When each arm could pass only one record ($k = 1$), ours gave 0 stale answers in 40 and the baseline gave

18 (45%), McNemar $p = 8 \times 10^{-6}$. The unchanged controls came out at 37 and 39 correct. We also tried $k = 2$ and $k = 5$ and got zero stale answers in both arms, apparently because the old and new versions arrive together and the model sorts them out by itself. Rank 1 decides the answer when only the top passage is kept, as in our one-line summaries.

F. System measurements

Speed was fine for a clinic. With everything running on the laptop, the median timeline query took 31.5 ms, and a chat question about one record came back in 35 ms because it never reaches the model. Writing a record takes 257 ms. Model-backed answers on the free tier take 2 to 3 s. The GPU only helps in bulk: MedCPT runs 12.9 times faster at batch 64 and no faster for one query. Over 250 automated backend tests pass, and so does a 16-step script that walks a real practice day against the live containers, including the HbA1c correction and an "as of 60 days ago" question.

VII. Limitations

We wrote the generator, and TAP-QA fixes the correction rate and ignores clinical severity, so absolute numbers will not carry over to real charts; the comparison between methods should hold up better. Every edge in TAP-QA is present and correct, which makes the graph partly a premise. We have not measured missing or wrong edges, and a real feed would have both.

All 972 corrections in seed 1 are laboratory values, and no chain is longer than one step. Longer chains and medication or diagnosis corrections are unit-tested but not benchmarked, and neither is the OCR path. Every arm shares one encoder, and the generation experiment used one hosted model, so the null result at $k \geq 2$ holds for that model only. Last, the method is simple on purpose. The parity between the hard filter and the weighted rule tells us the gain comes from having the supersession edge in the first place. That is the useful finding, but it also means we offer a formulation and a measurement, with no learned model.

VIII. Conclusion

A correction in a clinical chart withdraws an older entry, and neither entry's timestamps say so. When retrieval decides validity per record and per question date, computed over the pointer that clinical standards already store, stale answers disappear from TAP-QA and historical questions stay answerable, which a ranker with both timestamps cannot manage. For anyone building RAG over clinical records the advice is short: keep the correction pointer when you chunk and index, and resolve state at query time. Our next steps are a test on de-identified real charts with missing edges, longer correction chains, and a Hindi-capable encoder for code-switched notes.

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